DGMS 2019 • SECTION L: ENVIRONMENTAL GEOLOGY & WATER QUALITY

Question 89: Water Pollution & Organic Matter Loading: BOD vs. COD

Official Examination Guide • Biochemical Oxygen Demand, Streeter-Phelps Model & Dissolved Oxygen Deficit

OFFICIAL EXAMINATION QUESTION

The organic matter load in a water body is described as

(a) Dissolved oxygen	**(b) Biochemical oxygen demand** CORRECT ANSWER
(c) Chemical oxygen demand	(d) Total hardness

1. Executive Summary & Core Environmental Principle

In environmental geology, aquatic chemistry, and limnology (Masters & Ela, *Introduction to Environmental Engineering and Science*; Metcalf & Eddy, *Wastewater Engineering*):

- **Biochemical Oxygen Demand (BOD):** The primary parameter used to quantify the **biodegradable organic matter load** in natural water bodies (rivers, lakes, aquifers) and sewage effluents. It is defined as the mass of dissolved oxygen (O_2, expressed in mg/L) consumed by aerobic microorganisms during the biochemical oxidation and stabilization of organic carbon over a standardized incubation period (typically 5 days at 20°C, denoted as BOD_5):

> $\text{Organic Matter } (CH_2O) + O_2 \rightarrow [\text{Microorganisms}] \rightarrow CO_2 + H_2O + \text{Cellular Biomass}$

A high BOD value directly reflects a high concentration of decomposing organic matter (e.g., untreated sewage, agricultural runoff), which depletes dissolved oxygen (DO) and leads to aquatic hypoxia or fish mortality.

- **Chemical Oxygen Demand (COD):** Quantifies the total oxygen equivalent required to chemically oxidize all organic matter (both biologically degradable and biologically inert/refractory compounds) using a boiling mixture of concentrated sulfuric acid and a strong chemical oxidizing agent (potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$). For any given water sample, $\text{COD} \ge \text{BOD}$.
- **Dissolved Oxygen (DO):** The concentration of free, non-compound O_2 gas dissolved in water. It represents the oxygen resource available for aquatic fauna, which is consumed by the incoming BOD load.
- **Total Hardness:** The concentration of dissolved multivalent metallic cations, primarily calcium (Ca^{2+}) and magnesium (Mg^{2+}), expressed in mg/L as CaCO_3.

Therefore, the biodegradable organic matter load in a natural water body is universally described as **(b) Biochemical oxygen demand**.

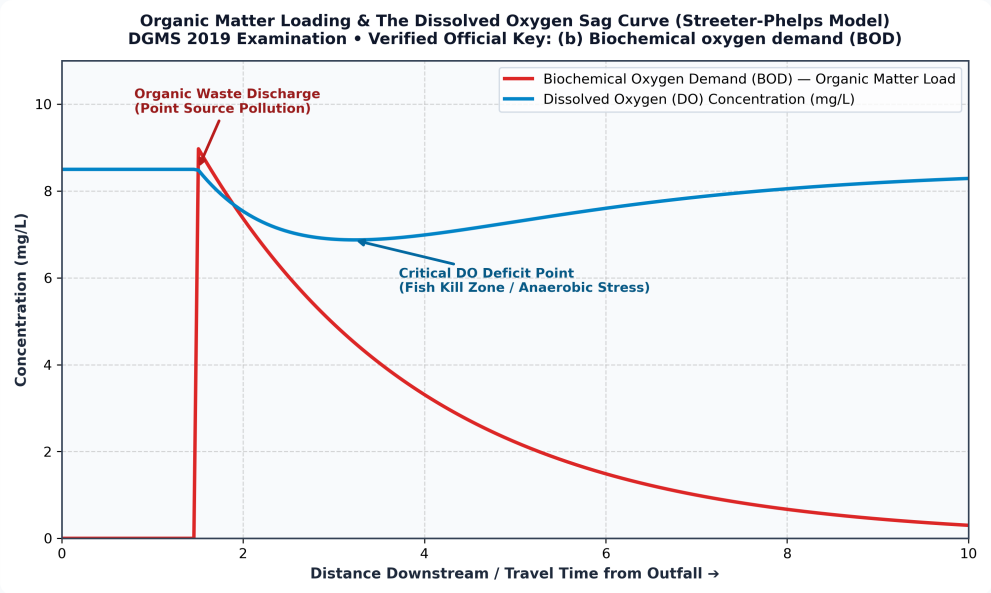


Figure 89.1: The classical Streeter-Phelps Oxygen Sag Curve illustrating how an organic pollution load (high BOD) triggers dissolved oxygen depletion (DO sag) in a receiving river system (Grounded in Masters & Ela).

2. Comparative Characterization of Water Quality Parameters

Parameter	Definition & Principle	Test Duration / Condition	Role in Water Quality Assessment
Biochemical Oxygen Demand (BOD)	O_2 consumed by aerobic bacteria decomposing biodegradable organics.	**5 days at 20°C (BOD_5)**; standard dilution method.	**Standard descriptor of organic pollution load** in natural rivers and lakes. Pristine water < 2 mg/L; polluted water > 10 mg/L.
Chemical Oxygen Demand (COD)	O_2 consumed by chemical digestion of all organics using boiling $\text{K}_2\text{Cr}_2\text{O}_7$ / H_2SO_4.	2 to 3 hours at 150°C.	Measures total oxidizable chemical and industrial waste load (biodegradable + refractory).
Dissolved Oxygen (DO)	Actual molecular oxygen dissolved in water.	Instantaneous electrochemical sensor or Winkler titration.	Critical for survival of fish and aerobic ecology (optimum 6–9 mg/L; lethal < 2 mg/L).
Total Hardness	Sum of $[\text{Ca}^{2+}]$ and $[\text{Mg}^{2+}]$	EDTA titration method.	Controls soap lathering and scaling in industrial boilers;

Parameter	Definition & Principle	Test Duration / Condition	Role in Water Quality Assessment
	polyvalent cations.		unrelated to organic matter load.

3. Evaluation of Options

- **(a) Dissolved oxygen: INCORRECT.** Represents the dissolved gas available for respiration, which is consumed by the organic load.
- **(b) Biochemical oxygen demand: CORRECT ANSWER.** By universal definition in environmental science, BOD quantifies the organic matter load in water bodies.
- **(c) Chemical oxygen demand: INCORRECT.** A laboratory chemical oxidation metric predominantly used for industrial wastewater, whereas the biological load in natural water bodies is characterized by BOD.
- **(d) Total hardness: INCORRECT.** Measures calcium and magnesium mineral content, not organic matter.

> **Key Takeaway:** Organic matter load in a water body = Biochemical Oxygen Demand (BOD). Hence, **(b) Biochemical oxygen demand** is the correct answer.

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